

A Ten-Paper Mass–Cancellation–Cover Audit:

Exact Closures, Independent Arithmetic Gates, and a Stop/Go Roadmap

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Abstract

TPC-83 through TPC-91 form nine consecutive audits of the fixed-shift matched-shell route; the present synthesis is the tenth paper in that block. We verify their exact dependency structure and separate what has been closed from what remains arithmetic.

On the magnitude side, TPC-83 proves the sharp equivalence between natural post-bin ℓ^1 mass and natural determinant-energy survival, while TPC-84 shows conditionally that any determinant-energy lower bound with fixed loss $\lambda_D < 1$ must be carried asymptotically by positive equal-determinant off-diagonal coherence. On the phase side, TPC-85 rules out three false normalization shortcuts. TPC-86 gives an exact coherent Fourier reconstruction and, in the zero-loss corner $\eta_Z = 0$ with a second-order filter, reduces the critical zero-mode input to a low-frequency window $R = X^{133/600+\varepsilon}$ for sufficiently small fixed $0 < \varepsilon < 107/240$; the general window depends on the desired η_Z and must remain below $q_X/2$. TPC-87 identifies the precise signed Cesàro defect needed to convert logarithmic cancellation to ordinary cancellation. TPC-88 factors determinant flatness as

$$\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}$$

and identifies the critical relative cancellation scale $\mathfrak{c} \leq X^{o(1)}/J$ in the natural-mass regime. TPC-89 proves that a surrogate transfers this scale uniformly only at relative ℓ^2 accuracy $O(J^{-1})$. TPC-90 completely classifies the primitive affine squarefree local support and isolates the remaining weighted Liouville sign correlation. TPC-91 models the stronger zero-residual full-block certificate as the constrained equation $Bc = w$, proves that bounded overlap does not imply it, and supplies projection and Farkas obstruction certificates. The original TPC-18 hypothesis also permits a separately controlled physical remainder, so exact coverage is a sufficient strengthening rather than the only admissible reassembly.

These are genuine L0 closures and L1 interfaces, but none is a new L2 estimate for the literal growing fixed- h_0 coefficient. The actual post-bin mass/energy lower bound and zero-mode upper bound remain open and their recorded exponents must satisfy $\lambda_D \leq 2\eta_Z$. TPC-18 full-block reassembly (by zero residual or a proved physical remainder), primitive determinant dispersion, and the strict physical loss $\Lambda_{\text{phys}} < 1/400$ remain independent. Equality $\lambda_D = 2\eta_Z$ is admissible but has no determinant reserve; $\Lambda_{\text{phys}} \geq 1/400$ stops the physical ledger. Losses are not counted twice.

We finish with a prioritized roadmap and explicit stop/go tests for four branches: low-frequency/log-Tauberian/surrogate zero-mode control, determinant mass and coherence, primitive affine sign arithmetic, and full-block reassembly certification. The audited outcome is forward but incomplete: no fixed-shift Hardy–Littlewood asymptotic, parity breach, prime-pair lower bound, or twin-prime conclusion is claimed.

Keywords: fixed shift; determinant energy; zero mode; Tauberian transfer; affine Möbius correlation; full-block cover; proof audit.

Literal-frame convention. Every physical statement retains one prescribed nonzero shift h_0 , the actual active support, all native keys, masks and prefixes, and one original global post-aggregation normalization. No result is specialized to $h_0 = 2$.

1 Introduction

Long proof programs accumulate two different kinds of progress. Some steps establish an arithmetic estimate on the literal physical coefficient. Others remove ambiguity: they identify the correct normalization, prove an exact transfer theorem, or exhibit a counterexample to an invalid shortcut. Both are useful, but they must not be reported at the same proof level.

TPC-83 through TPC-91 were designed as a controlled exploration of the fixed-shift determinant route after the completion-to-flatness audit in TPC-82 [2]. They investigate three quantities and two reassembly interfaces:

$$M_X = \sum_n |A_X(n)|, \quad D_X = \sum_n |A_X(n)|^2, \quad Z_X = \sum_n A_X(n),$$

together with the TPC-18 all-block reassembly and primitive determinant dispersion [1]. Here A_X is always the already aggregated literal determinant coefficient on its actual support. At the audited critical row we use

$$Q_X = X^{267/400+o(1)}, \quad J_X = X^{133/400+o(1)}, \quad N_{0,X} = J_X Q_X^2.$$

The padded Fourier modulus satisfies $q_X \asymp Q_X$, and

$$d_q(r, 0) = \min_{m \in \mathbb{Z}} |r + mq|$$

denotes cyclic distance. If $v \neq 0$ is a coefficient on the same literal frame Ω , with $S = \#\Omega$, put

$$Z(v) = \sum_{\omega \in \Omega} v_\omega, \quad D(v) = \sum_{\omega \in \Omega} |v_\omega|^2, \quad \rho(v) = \frac{|Z(v)|}{\sqrt{SD(v)}}.$$

Every surrogate comparison below uses this same Ω ; zero padding or changing the frame changes ρ .

The nine papers do not supply nine interchangeable estimates. Instead, they form four branches:

- (i) post-bin mass, determinant energy, and equal-determinant coherence;
- (ii) distinguished zero-mode cancellation through Fourier, Tauberian, or surrogate interfaces;
- (iii) primitive affine local support versus the remaining sign correlation; and
- (iv) full-block reassembly of the upstream hard packet, with exact coverage as its zero-residual sufficient certificate.

These branches must merge only after their normalizations and losses have been matched literally.

1.1 Purpose of the present audit

This paper has four tasks. First, it records the precise dependency graph for TPC-83–TPC-91 from their actual theorem statements, rather than from suggestive titles. Second, it classifies every advance as L0, L1, or L2. Third, it identifies which routes have been sharpened, which shortcuts have been killed, and which arithmetic gates remain independent. Fourth, it turns those gates into a prioritized roadmap with explicit stop/go criteria.

The result is not a proof by accumulation. Exact identities do not become a growing fixed-shift correlation estimate merely because many of them are now available. Conversely, a negative transfer result can be durable progress when it prevents an invalid route from being retried.

1.2 Headline conclusion

The nine-paper block has made the route substantially more falsifiable:

- the correct mass variable is post-bin, not raw;
- any determinant-energy lower bound with fixed loss below one requires equal-determinant coherence;
- the zero mode is a coherent low-frequency or affine-sign problem, not a centering convention;
- logarithmic-to-ordinary transfer requires a signed Tauberian defect with a rate that survives outer reassembly;
- a surrogate must have uniform worst-case accuracy at the full J^{-1} scale (or $X^{o(1)}/J_X$ in the physical convention);
- local squarefree density does not supply sign cancellation; and
- bounded overlap does not supply an exact full-block certificate, while a nonzero coefficient residual still needs a physical remainder theorem.

No paper in the block proves the remaining growing arithmetic bound. The honest aggregate status is therefore strong L0/L1 progress with no new L2 theorem.

1.3 Organization

[Section 2](#) fixes the proof levels and literal invariants. [Section 3](#) gives the dependency graph and paper-by-paper scorecard. [Section 4](#) records the exact positive advances and killed shortcuts. [Section 5](#) states the remaining conjunction and separate ledgers. [Sections 6](#) and [7](#) give the branch tests and priority order. [Section 8](#) states the final audited boundary.

2 Audit protocol and invariant frame

2.1 Proof levels

Definition 2.1 (Three proof levels). We use the following noninterchangeable labels.

L0: finite model or abstract theorem.

An exact algebraic, geometric, Fourier, convex, or Tauberian theorem, including a finite or structure-free countermodel.

L1: literal interface theorem.

An exact transfer that identifies the L0 quantity on the physical carrier while preserving every required key, mask, support, coefficient, and normalization. Its hypotheses may still include an unproved arithmetic estimate.

L2: growing arithmetic estimate.

A new uniform bound for the actual growing fixed- h_0 physical coefficient, with all prefixes, exceptional sectors, polarizations, and outer reassembly returned.

Remark 2.2. An averaged shift, almost-all scale, fixed-form, frozen-family, support-only, or positive-square estimate is recorded at its native scope. It is not promoted to L2 fixed- h_0 progress without a proved literal transfer.

2.2 The literal invariant tuple

Definition 2.3 (Literal invariant frame). At scale X , the physical frame consists of:

$$\mathfrak{I}_X = (h_0, \Omega_X, \mathcal{K}_X, \mathcal{W}_X, \mathcal{N}_X).$$

Here $h_0 \neq 0$ is the prescribed shift, Ω_X is the actual active atom or determinant support, $\mathcal{K}_X$ is the complete native-key and provenance list, $\mathcal{W}_X$ contains all masks, weights, prefixes, interval origins, and polarizations, and $\mathcal{N}_X$ is the single global post-aggregation normalizer.

A transfer is literal only when it gives an exact map between the source and target frames. Equal numerical determinants do not justify merging distinct native atoms. Zero padding, changing a fiber mass, or replacing the post-bin mass by a raw majorant changes the relevant statistic and requires a separate theorem.

Proposition 2.4 (Promotion rule). *Within this audit, an L0 or conditional L1 statement is promoted to an L2 statement only after a theorem verifies:*

- (a) *the same literal frame $\mathfrak{I}_X$;*
- (b) *uniformity in every growing physical coefficient and prefix;*
- (c) *return of every exceptional, resonant, short-fiber, and outer-reassembly sector; and*
- (d) *the exact endpoint loss in the correct ledger.*

Proof. These items are part of the definition of the audited L2 target. Omitting one changes either the coefficient, its quantifiers, or its normalization, so it proves a different statement. $\square$

2.3 What counts as forward progress

An audited step is forward if it does at least one of the following:

- (i) proves a new literal fixed-power or logarithmic estimate;
- (ii) closes a necessary exact interface in the dependency graph;
- (iii) reduces a required estimate to a strictly smaller literal family with controlled return; or
- (iv) rules out a proposed route by an exact physical counterexample or dual obstruction.

Numerical stability, a pilot census, or a phase-free upper bound may guide the next experiment, but is not by itself any of these four items.

3 Dependency graph and paper-by-paper scorecard

3.1 Directed dependency graph

The arrows below mean “uses or sharpens an interface from,” not “logically implies the final arithmetic theorem”:

$$\begin{array}{ll} \text{TPC-83} \longrightarrow \text{TPC-84}, & \text{TPC-83} \longrightarrow \text{TPC-88} \longrightarrow \text{TPC-89}, \\ \text{TPC-85} \longrightarrow \text{TPC-86} \longrightarrow \text{TPC-87}, & \text{TPC-85, TPC-87} \longrightarrow \text{TPC-90}, \\ \text{TPC-18} \longrightarrow \text{TPC-91}, & \text{TPC-83-91} \longrightarrow \text{TPC-92}. \end{array}$$

TPC-84 is the coherence continuation of the mass branch. TPC-86 and TPC-87 sharpen the Fourier and logarithmic alternatives opened by TPC-85. TPC-88 and TPC-89 express the final flatness target as relative phase cancellation and then as a sharp surrogate-angle requirement. TPC-90 opens the affine correlation from TPC-85 after the Tauberian warning in TPC-87. TPC-91 is a cross-cutting audit of the separate TPC-18 reassembly gate.

3.2 Exact scorecard

Theorem 3.1 (Dependency audit). *The outputs summarized in table 1 close exact identities, sharp stability thresholds, and falsifiable interfaces. Taken together, they do not prove any of the remaining-gate entries in the same row or in another row.*

Proof. TPC-83 and TPC-88 explicitly assume rather than prove mass or energy survival [3, 8]. TPC-84 proves only a conditional necessity for positive coherence [4]. TPC-85–TPC-87 give identities, reductions, and countermodels but no literal Möbius bound [5–7]. TPC-89 supplies a sharp transfer conditional on a surrogate that it does not construct [9]. TPC-90 proves local squarefree support while leaving the Liouville product unresolved [10]. TPC-91 defines and diagnoses the stronger zero-residual certificate and its residual alternative, but neither instantiates the full physical system, controls its physical remainder, nor proves determinant dispersion [11]. Therefore no remaining-gate entry is supplied by the audited theorem statements. $\square$

Paper	Exact advance	Remaining literal gate	Highest new level
TPC-83	$M^2/S < D \leq LM$; natural post-bin mass $\iff$ natural energy	Prove a lower bound for the actual post-bin M_X or D_X	L0 + L1
TPC-84	Sharp fiber resultant interval; $D = E_{\text{trip}} + C_{\text{eq}}$	Prove dominant-native-term survival on enough actual fibers, or a positive same-scale structured C_{eq} with controlled remainder; census not executed	L0 + L1
TPC-85	Exact zero-mode disintegration and three normalization barriers	Bound the literal Z_X , coherent modes, or growing affine pair	L0 + L1
TPC-86	Optimal zero-set kernel and low-frequency reduction to $R = X^{133/600+\varepsilon}$ when $\eta_Z = 0$, $k = 2$, and $0 < \varepsilon < 107/240$	Prove the multiplier-weighted literal low-frequency estimate	L0 + L1
TPC-87	Exact signed Cesàro criterion and sharp structure-free countermodel	Uniform terminal cancellation for all physical affine fibers and prefixes	L0 + L1
TPC-88	$\mathfrak{F} = \mathfrak{c}^2 S_{\text{eff}}$; critical relative target $X^{o(1)}/J$	Prove natural mass and the required relative phase cancellation	L0 + L1
TPC-89	Sharp principal-angle perturbation; accuracy threshold $O(J^{-1})$	Construct and prove a literal same-frame surrogate at that accuracy	L0 + L1
TPC-90	Exact local roots, uniformly positive local Euler product, fixed-pair squarefree density, and $\mu^2\lambda$ sign split	Growing fixed- h_0 weighted Liouville cancellation with full return	L0 + L1 interface
TPC-91	Exact zero-residual $Bc = w$ certificate, projection residual, and Farkas obstruction	Build the physical incidence system; prove exact cover or control its physical residual; separately prove TPC-18 determinant dispersion	L0 + L1 schema

Table 1: No row contains a new L2 growing fixed-shift arithmetic estimate.

4 Exact advances and permanently removed shortcuts

4.1 Magnitude branch: mass, energy, and coherence

Let

$$S_X = \# \text{supp } A_X, \quad L_X = \|A_X\|_\infty, \quad M_X = \|A_X\|_1, \quad D_X = \|A_X\|_2^2,$$

and $N_{0,X} = J_X Q_X^2$. TPC-83 combines the universal inequalities

$$\frac{M_X^2}{S_X} \leq D_X \leq L_X M_X$$

with the verified literal envelopes

$$S_X \ll Q_X, \quad L_X \ll X^{o(1)} \frac{N_{0,X}}{Q_X}.$$

It follows exactly that subpower natural mass survival is equivalent to subpower natural energy survival. This closes the choice of variable, not its lower bound [3].

TPC-84 opens the post-bin coefficient into native determinant fibers. For fiber moduli r_i , the sharp free-phase resultant interval is

$$\left[(2r_{\max} - \sum_i r_i)_+, \sum_i r_i \right].$$

It also proves

$$D_X = E_{\text{trip},X} + C_{\text{eq},X},$$

where the physical triplet diagonal is at most on the loss-one scale. Thus an eventual energy lower bound with fixed loss $\lambda_D < 1$ must be carried asymptotically by positive equal-determinant off-diagonal coherence. This is a necessary mechanism, not a proved lower bound [4].

Proposition 4.1 (Audited magnitude output). *Under the literal support and pointwise envelopes above, natural post-bin mass survival is equivalent to natural determinant-energy survival. Moreover, if a future theorem proves*

$$D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2 \quad \text{with fixed } \lambda_D < 1,$$

then

$$C_{\text{eq},X} = (1 - o(1)) D_X > 0.$$

Neither conclusion supplies the mass or energy lower bound itself.

Proof. The first statement is the two-sided TPC-83 exponent transfer. For the second, TPC-84 proves that the physical triplet diagonal is at most on the loss-one scale, so it is $o(D_X)$ under the displayed fixed-loss hypothesis. Insert this in $D_X = E_{\text{trip},X} + C_{\text{eq},X}$. $\square$

A continuation may seek dominant-native-term survival on enough actual fibers, or a positive structured same-determinant pairing with controlled remainder. These are research strategies, not an exhaustive theorem-level disjunction. Raw mass precedes fiber summation and cannot lower-bound the post-bin norm.

4.2 Zero-mode branch: Fourier, Tauberian, and surrogate interfaces

TPC-85 proves that

$$A_X(0) = 0 \quad \not\Rightarrow \quad Z_X = \hat{A}_X(0) = 0.$$

Under no-wrap padding,

$$A_X = A_X^\circ + \frac{Z_X}{q_X} \mathbf{1}, \quad D_X = \|A_X^\circ\|_2^2 + \frac{|Z_X|^2}{q_X},$$

and

$$Z_X = - \sum_{r \neq 0} \hat{A}_X(r).$$

Conditional expectation preserves Z_X and changes the energy normalization; content inversion returns the full shell in its $t = 1$ term. These identities eliminate three normalization shortcuts but retain coherent Fourier reassembly as a valid branch [5].

TPC-86 sharpens that branch. A positive-density structural zero arc supports a fixed-order filter with bounded Wiener norm. For the literal packet,

$$|Z_X| \ll_k \max_{0 < d_{q_X}(r,0) \leq R} |\hat{A}_X(r)| + X^{o(1)} N_{0,X} R^{1/2-k}.$$

Write

$$R_X = X^{\varrho+o(1)} < q_X/2, \quad \varrho < \frac{267}{400}.$$

At the critical row, a target

$$|Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2$$

has an admissible filter tail whenever

$$(k - \tfrac{1}{2})\varrho > \frac{133}{400} + \eta_Z.$$

The often quoted choice

$$k = 2, \quad R_X = X^{133/600+\varepsilon}$$

is the zero-loss corner $\eta_Z = 0$, with sufficiently small fixed

$$0 < \varepsilon < \frac{267}{400} - \frac{133}{600} = \frac{107}{240}.$$

More generally that same window requires $\eta_Z < 3\varepsilon/2$. The missing input is therefore a coherent literal low-frequency estimate at a window matched to the desired rate, not control of all q_X frequencies [6].

TPC-87 identifies the exact cost of importing logarithmic cancellation. For

$$S_N = \sum_{n \leq N} a_n, \quad L_N = \sum_{n \leq N} \frac{a_n}{n},$$

one has

$$\frac{S_N}{N} = L_N - \underbrace{\frac{1}{N} \sum_{m < N} L_m}_{\mathcal{T}_N(L)}.$$

Thus ordinary cancellation is exactly the vanishing of the signed backward Cesàro defect. Normalized logarithmic cancellation alone does not imply it. Terminal-half control or vanishing dyadic logarithmic variation can bound the defect for one sequence, but a physical application must also survive the weighted outer sum. If

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} S_{\theta,X}(N_{\theta,X}) + E_{\text{content},X},$$

the relevant quantitative certificate is

$$\sum_{\theta \in \Theta_X} |c_{\theta,X}| N_{\theta,X} \left| \mathcal{T}_{N_{\theta,X}}(L_{\theta,X}) \right| + |E_{\text{content},X}| \leq X^{-\eta_Z+o(1)} Q_X^2. \quad (1)$$

The Abel identity itself is L0; the growing estimate (1) is not proved [7].

TPC-88 and TPC-89 turn the same target into magnitude–phase and projective geometry. For $A_X \neq 0$,

$$\mathfrak{F}_X = \frac{|Z_X|^2}{D_X} = \underbrace{\left(\frac{|Z_X|}{M_X} \right)^2}_{c_X^2} \underbrace{\frac{M_X^2}{D_X}}_{S_{\text{eff},X}}.$$

Under natural post-bin mass survival,

$$S_{\text{eff},X} = X^{o(1)} Q_X,$$

and at the critical endpoint

$$\mathfrak{F}_X \leq X^{1/400+o(1)} \iff \mathfrak{c}_X \leq \frac{X^{o(1)}}{J_X} \iff |Z_X| \leq X^{o(1)} Q_X^2.$$

If a same-frame surrogate b_X is used, its projective angle transfers with the sharp law

$$|\rho(A_X) - \rho(b_X)| \leq \frac{\|A_X - b_X\|_2}{\|b_X\|_2}.$$

Uniform transfer of the critical angle therefore requires relative accuracy $O(J_X^{-1})$ in order; $o(1)$ is insufficient [8, 9].

Theorem 4.2 (Audited zero-mode fork). *In the natural-mass corner, any positive zero-mode continuation must establish at least one of the following on the same literal frame:*

- (i) *the TPC-86 coherent low-frequency bound with $(k - \frac{1}{2})\varrho > 133/400 + \eta_Z$;*
- (ii) *the TPC-87 signed Tauberian defects on every required affine fiber and prefix, with the aggregate rate (1);*
- (iii) *a same-frame TPC-89 surrogate with both angle and relative error $X^{o(1)}/J_X$; or*
- (iv) *a direct bound of equivalent strength for Z_X .*

The audited results prove the equivalence or sufficiency of these certificates where stated, but prove none for the growing physical coefficient.

Proof. Items (i)–(iii) are the literal sufficient interfaces proved in TPC-86, TPC-87, and TPC-89. TPC-88 converts their common target to the displayed Z_X scale under natural mass. Each source paper explicitly leaves its growing arithmetic hypothesis unproved. $\square$

4.3 Primitive affine branch

For individually primitive forms

$$L_1(r) = d + sr, \quad L_2(r) = u + ar, \quad su - ad = h_0 \neq 0,$$

TPC-90 proves

$$\gcd(a, s) \mid h_0, \quad \gcd(L_1(r), L_2(r)) \mid h_0.$$

If $\nu_{p,k}$ counts residues modulo p^k on which $p^k \mid L_1(r)$ or $p^k \mid L_2(r)$, then

$$\nu_{p,k} = \mathbf{1}_{p \nmid s} + \mathbf{1}_{p \nmid a} - \mathbf{1}_{p \nmid as} \mathbf{1}_{p^k \mid h_0}.$$

Consequently the simultaneous squarefree local product is positive and uniformly bounded below by

$$\prod_p \left(1 - \frac{2}{p^2}\right) > 0.$$

On positive intervals the exact sign split is

$$\mu(L_1)\mu(L_2) = \mu^2(L_1)\mu^2(L_2)\lambda(L_1)\lambda(L_2).$$

All square-power local obstructions are now explicit; the unresolved quantity is the weighted Liouville product on the actual growing physical family [10].

4.4 Full-block reassembly branch

TPC-91 places the zero-residual strengthening of the TPC-18 all-block hypothesis on the literal atom space. With exact candidate columns B_X , target w_X , and the coefficient rule $\mathcal{C}_X$ actually authorized upstream, this certificate is

$$B_X c_X = w_X, \quad c_X \in \mathcal{C}_X.$$

For unrestricted synthesis the exact defect is

$$\varepsilon_{\text{cov}} = \|(I - B_X B_X^\dagger) w_X\|_2.$$

For nonnegative synthesis, Farkas' alternative gives an exact dual obstruction. For a general coefficient rule, exact feasibility is the membership $w_X \in B_X \mathcal{C}_X$; zero distance is equivalent only when the attainable image is closed. Bounded overlap, even overlap one with every atom covered, need not give an $o(1)$ -relative coefficient cover.

The original TPC-18 hypothesis is weaker: it may allow

$$w_X = B_X c_X + r_X, \quad \mathcal{T}_X(r_X) = o(X)$$

in its physical scale, where $\mathcal{T}_X$ denotes the original literal physical reassembly functional, with the prescribed coefficients and every quantitative loss retained. Thus exact coverage is the clean zero-remainder sufficient route. Its failure does not kill TPC-18 reassembly unless the physical residual route is also ruled out [1, 11].

4.5 Shortcuts removed by exact counterexamples

Invalid shortcut	Exact reason it fails	Source
Raw or pre-bin absolute mass $\Rightarrow$ post-bin mass	Opposite atoms may cancel inside one determinant fiber	TPC-83/84
Magnitude spreading $\Rightarrow$ phase cancellation	Identical magnitudes can have maximal or zero flatness	TPC-88
$A_X(0) = 0$, centering, conditional expectation, or content inversion $\Rightarrow Z_X = 0$	Each operation has an exact identity that retains or extracts the unknown zero mode	TPC-85
Mean-square or all-but-exceptional frequencies $\Rightarrow$ coherent reassembly	Exceptional modes must be charged by multiplier mass and phase	TPC-86
Normalized logarithmic cancellation $\Rightarrow$ ordinary mean	The dyadic countermodel has terminal log cancellation but ordinary limits $\pm 1/3$	TPC-87
Relative surrogate error $o(1) \Rightarrow J^{-1}$ angle control	The sharp tangent example has angle error exactly δ	TPC-89
Positive squarefree density $\Rightarrow$ sign cancellation	The remaining factor is the independent Liouville product	TPC-90
Bounded overlap, coarse-key cover, or small Euclidean residual $\Rightarrow$ exact or physically controlled reassembly	Explicit residual and Farkas counterexamples prevent exact feasibility; Euclidean size alone does not control the physical image	TPC-91

Table 2: These routes are removed as automatic transfers. A new theorem using additional arithmetic structure is not excluded.

5 The remaining independent gates

5.1 Current-route completion vector

The exact papers now permit the remaining hypotheses to be stated without substitution. Fix exponent parameters $\lambda_D, \eta_Z \geq 0$. On the current route, write:

$$\begin{aligned} G_{\text{mass}} : \quad & D_X \geq X^{-\lambda_D + o(1)} Q_X^3 J_X^2, \\ G_{\text{zero}} : \quad & |Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2, \\ G_{\text{reasm}} : \quad & w_X = B_X c_X + r_X, \quad c_X \in \mathcal{C}_X^{\text{up}}, \quad \mathcal{T}_X(r_X) = o(X), \\ G_{\text{DD}} : \quad & \text{the TPC-18 primitive determinant estimate holds on every returned block,} \\ G_{\text{phys}} : \quad & \Lambda_{\text{phys}} < \frac{1}{400}. \end{aligned}$$

Here $\mathcal{C}_X^{\text{up}}$ is the coefficient rule actually derived upstream; in the original unit-coefficient packet it is not replaceable by an arbitrary signed solution. The reassembly gate includes coefficient amplification and the quantitative residual rate required by the physical ledger, and $\mathcal{T}_X$ is the original literal physical reassembly functional. The exact cover $B_X c_X = w_X$ is the special sufficient case $r_X = 0$. The determinant-dispersion gate includes every prefix and exceptional block required by TPC-18.

Definition 5.1 (Audited completion conjunction). The current route has an *audited completion certificate* only when all five gates above hold in one literal frame and the recorded mass/zero-mode bounds satisfy

$$\lambda_D \leq 2\eta_Z. \quad (2)$$

This definition does not assert that the gates have been proved. It records the conjunction required by the present chain. A genuinely different proof may bypass a gate, but must then replace the corresponding dependency by another literal theorem.

5.2 Determinant and physical ledgers

The determinant compatibility (2) compares the recorded lower bound for D_X with the recorded upper bound for Z_X . At the critical row it is the sufficient flatness ledger derived in TPC-82 and reproduced exactly by TPC-88 [2, 8]. If $\lambda_D > 2\eta_Z$, those particular recorded bounds do not certify the target. This failure need not disprove a sharper bound for the actual coefficient.

The physical ledger records fixed-power degradation elsewhere in the literal interface and downstream comparison:

$$\boxed{\Lambda_{\text{phys}} < \frac{1}{400}}. \quad (3)$$

It is strict. Equality or excess stops the route. A polynomial reassembly-amplification or physical-residual loss enters this ledger once if it is not already present in the block error. A determinant-energy loss already represented by λ_D , or a zero-mode gain represented by η_Z , is not charged again.

Theorem 5.2 (Ledger separation). *The audited chain requires the conjunction*

$$\lambda_D \leq 2\eta_Z \quad \text{and} \quad \Lambda_{\text{phys}} < 1/400.$$

It supplies no theorem that adds these inequalities into one budget, converts one into the other, or permits the same loss to be entered twice.

Proof. The first inequality compares two statistics of the same literal post-bin coefficient. The second counts losses in distinct physical maps and reassembly steps, excluding losses already represented in the first comparison. They have different definitions and inputs. TPC-87–TPC-91 explicitly preserve this separation [7–11]. $\square$

Remark 5.3 (Equality). Equality $\lambda_D = 2\eta_Z$ is compatible with the recorded flatness exponent but leaves no determinant-ledger reserve. Equality $\Lambda_{\text{phys}} = 1/400$ violates the strict physical rule and is a stop. These two equality cases must not be conflated.

5.3 Reassembly and determinant dispersion

Proposition 5.4 (TPC-18 gate independence). *Within the audited theorem chain:*

- (i) *an exact allowed cover $B_X c_X = w_X$ is a sufficient zero-residual reassembly certificate, but does not prove the primitive determinant estimate on any column;*
- (ii) *a determinant estimate on every candidate column does not prove either exact coverage or a physical bound for the residual;*
- (iii) *a nonzero coefficient residual does not by itself fail the TPC-18 reassembly gate, but closes it only after $\mathcal{T}_X(r_X)$ is bounded at the required rate.*

Proof. Zero-residual coverage is a constrained coefficient identity. Determinant dispersion is an arithmetic bound on the column statistics. TPC-91 gives finite systems with bounded overlap but positive cover defect, while the TPC-18 estimate is stated conditionally for already admissible blocks [1, 11]. If $w_X = B_X c_X + r_X$, reassembly contains the additional term $\mathcal{T}_X(r_X)$; Euclidean smallness alone does not estimate it. $\square$

5.4 Local support and sign arithmetic

The TPC-90 product

$$\prod_p (1 - \nu_{p,2}/p^2) > 0$$

certifies that primitive affine squarefree support has no local annihilation. It is not a saving in η_Z , a lower bound for D_X , a TPC-18 determinant estimate, or a reduction of Λ_{phys} . The remaining Liouville product is a signed arithmetic gate and must be estimated on its literal growing family.

Gate	Required literal certificate	Present status
Mass/energy	G_{mass} , equivalently natural post-bin mass in the subpower corner	Transfer proved; lower bound open
Zero mode	G_{zero} , via Fourier, Tauberian, surrogate, or direct estimate	Exact reductions proved; growing estimate open
Full-block reassembly	Upstream-prescribed $w_X = B_X c_X + r_X$, with exact cover or a physical bound on $\mathcal{T}_X(r_X)$	Exact-cover certificate theory proved; physical system and residual route not solved
Primitive dispersion	Uniform TPC-18 estimate on every returned block	Independent conditional input
Endpoint	$\lambda_D \leq 2\eta_Z$ and $\Lambda_{\text{phys}} < 1/400$, with no duplication	Bookkeeping exact; hypotheses unproved

Table 3: No row may be substituted for another.

6 Explicit branch stop/go tests

A stop test kills only the stated branch or certificate unless an end-to-end impossibility theorem is proved. Numerical failure is a diagnostic, not a stop certificate.

6.1 Branch I: zero-mode control

Fourier go test. Fix the TPC-86 filter of order $k \geq 2$, its literal padding, and

$$R_X = X^{\varrho+o(1)} < q_X/2, \quad \varrho < \frac{267}{400}, \quad (k - \tfrac{1}{2})\varrho > \frac{133}{400} + \eta_Z.$$

For the zero-loss target $\eta_Z = 0$, the second-order choice $R_X = X^{133/600+\varepsilon}$ is admissible for fixed $0 < \varepsilon < 107/240$; for positive η_Z the window must be enlarged according to the displayed condition

while remaining below $q_X/2$. The branch advances if the multiplier-weighted low-frequency term and returned tail jointly imply

$$|Z_X| \leq X^{-\eta_Z + o(1)} Q_X^2$$

for an η_Z compatible with (2). Every exceptional frequency must be charged by reconstruction-multiplier mass, not cardinality.

Fourier stop test. The particular filter certificate stops if a rigorous literal lower bound on its exceptional or resonant contribution makes the target impossible, if the zero arc loses positive density without a replacement estimate, or if its unavoidable physical loss makes $\Lambda_{\text{phys}} \geq 1/400$. A mean-square theorem or an all-but-exceptional theorem that does not return the weighted exceptional set is merely insufficient input, not a global impossibility result.

Tauberian go test. For every physical affine sequence and every prefix used by outer reassembly, prove a quantitative signed defect, terminal-half, or variation bound such that the complete returned estimate satisfies

$$\sum_{\theta \in \Theta_X} |c_{\theta,X}| N_{\theta,X} \left| L_{\theta,X}(N_{\theta,X}) - \frac{1}{N_{\theta,X}} \sum_{m < N_{\theta,X}} L_{\theta,X}(m) \right| + |E_{\text{content},X}| \leq X^{-\eta_Z + o(1)} Q_X^2,$$

or the corresponding quantitative target for the literal block to which the certificate is attached. A per-fiber $o(1)$ statement without this outer rate does not pass. The reciprocal weight introduced on translated fibers must be transferred exactly.

Tauberian stop test. A theorem controlling only normalized logarithmic means, only fixed forms, or almost all scales/origins fails this certificate until its exceptional quantifiers are removed. A proof that no available uniformity can survive the literal growing coefficients kills that transfer route, not Möbius-specific alternatives.

Surrogate go test. Construct a same-frame b_X and prove

$$\rho(b_X) \leq \frac{X^{o(1)}}{J_X}, \quad \frac{\|A_X - b_X\|_2}{\|b_X\|_2} \leq \frac{X^{o(1)}}{J_X}.$$

The support map must be exact because zero padding changes ρ .

Surrogate stop test. The chosen surrogate class is killed by a rigorous lower bound showing that every allowed b_X has relative error $\gg X^c/J_X$ for some fixed $c > 0$, or by an exact nonidentifiability counterexample. Merely obtaining $o(1)$ error does not pass the gate.

6.2 Branch II: determinant mass and coherence

Mass go test. Prove

$$M_X \geq X^{-o(1)} N_{0,X}$$

for the literal post-bin coefficient, or prove the equivalent natural energy lower bound directly. A more general fixed-loss result is still usable, but must enter the full (μ, λ_D, η_Z) profile rather than the natural corner.

Census go test. Execute the native-key TPC-84 decomposition with exact determinant keys, then prove either a dominant-term lower bound on enough fibers or a positive structured pairing whose unpaired remainder is smaller uniformly in X . Finite computation can certify finite identities; the growing lower bound requires a theorem.

Mass stop tests. The phase-blind dominant-term strategy stops if its exact lower envelope is too small. The natural-mass branch stops if one proves on an unbounded sequence of scales the literal upper bound

$$M_X \leq X^{-c+o(1)} N_{0,X}$$

for some fixed $c > 0$. Neither event rules out a nonnatural loss-profile route. A raw majorant or unexecuted census proves no stop.

6.3 Branch III: primitive affine sign arithmetic

Go test. Prove cancellation for

$$\sum_{\substack{r \\ L_{1,\theta,X}(r) L_{2,\theta,X}(r) \neq 0}} W_{\theta,X}(r) \mu^2(|L_{1,\theta,X}(r)|) \mu^2(|L_{2,\theta,X}(r)|) \lambda(|L_{1,\theta,X}(r)|) \lambda(|L_{2,\theta,X}(r)|)$$

uniformly over the literal growing coefficients, prescribed nonzero determinant h_0 , origins, masks, relevant prefixes, short-fiber treatment, polarizations, and outer keys. The aggregate bound must fit either the zero-mode target or the TPC-18 block target to which it is attached.

Weaker go tests. A fixed-power saving on one literal residual block, a certified range of square-divisor moduli, or a smoothed/frozen one-slot theorem is publishable progress at its native level if it has an exact route back toward the physical coefficient. It is not reported as full fixed- h_0 cancellation.

Stop tests. Further local Euler-product calculations do not address the remaining sign factor. The present implementation stops if a rigorous transfer lower bound shows that returning its excluded sectors necessarily forces $\Lambda_{\text{phys}} \geq 1/400$. An averaged-shift or fixed-form theorem that cannot select the prescribed growing family is an input mismatch, not proof that every affine route fails.

6.4 Branch IV: full-block reassembly

Exact-cover go test. Export the complete physical target and candidate blocks to one native row list. Solve the coefficient rule actually derived upstream; for the original unit cover the test is

$$B_X \mathbf{1} = w_X,$$

not merely existence of an arbitrary signed solution. Verify the coefficient-weighted block error under the original normalization.

Exact-cover stop test. For the proposed block family, any verified nonzero orthogonal residual or applicable Farkas vector is a rigorous stop certificate for that zero-residual candidate family. It is not a stop for the original TPC-18 asymptotic reassembly. A coarse determinant-key solution does not pass.

Physical-residual go test. If only $w_X = B_X c_X + r_X$ is available with the upstream-prescribed coefficients, prove the required physical estimate for $\mathcal{T}_X(r_X)$, retain its rate in Λ_{phys} , and verify the original global normalization. Small $\|r_X\|_2$ alone is not this theorem.

Determinant-dispersion go test. After an admissible exact or residual reassembly, prove the TPC-18 primitive generic estimate uniformly on every returned block with the required strict exponent margin. Reassembly and dispersion are checked separately.

Global stop test. Even if every local branch passes, equality or excess in Λ_{phys} stops the assembled route. Failure of $\lambda_D \leq 2\eta_Z$ stops only the certificate formed from those recorded D_X, Z_X bounds unless a sharper estimate is impossible.

Branch	Go certificate	Rigorous branch kill
Zero mode	Literal low-frequency, signed-Tauberian, surrogate, or direct Z_X bound	Unavoidable weighted residual or accuracy lower bound beyond the target
Mass/coherence	Post-bin M_X or D_X lower bound; structured coherence with remainder	Literal fixed-power mass-loss upper bound kills the natural corner
Affine sign	Growing fixed- h_0 weighted Liouville cancellation with full return	Proved return loss reaches $1/400$, or exact obstruction for the specified subroute
Full-block reassembly	Allowed exact $B_X c_X = w_X$, or a proved physical residual; then separate DD	Nonzero dual/Farkas residual kills only that exact-cover family

Table 4: A failed heuristic or finite experiment is not a rigorous kill certificate.

7 Prioritized next-stage roadmap

The priorities below are ordered by how directly the existing exact reductions expose a checkable next theorem. Priorities one and two may run in parallel because zero-mode cancellation and energy survival are both required.

7.1 Priority 1: zero-mode low-frequency, Tauberian, and surrogate branch

Deliverable 1A: literal low-window export. Write every $\hat{A}_X(r)$, $0 < d_{q_X}(r, 0) \leq R_X$, in the native affine variables, retain the TPC-86 multiplier, and partition resonant, short-fiber, and generic sectors without deletion. The first theorem target is the complete multiplier-weighted window, not an individual generic frequency. Choose $R_X = X^{\varrho+o(1)}$ and filter order k so that $R_X < q_X/2$, $\varrho < 267/400$, and $(k - \frac{1}{2})\varrho > 133/400 + \eta_Z$; the $133/600 + \varepsilon$ window, with fixed $0 < \varepsilon < 107/240$, is reserved for the second-order zero-loss corner unless $\eta_Z < 3\varepsilon/2$.

Deliverable 1B: signed Tauberian certificate. For any logarithmic input, compute the exact unnormalized primitive and its signed backward Cesàro defect on every physical prefix. Then prove that the coefficient-weighted fiber defects, short-fiber return, and content completion together meet (1), or the corresponding quantitative target of the attached block. A per-fiber $o(1)$ defect is not enough when the number or weights of fibers grow. If only normalized window cancellation is available, record a negative transfer result rather than treating it as ordinary cancellation.

Deliverable 1C: surrogate feasibility test. Construct candidate same-frame surrogates only after an exact support map. Prove or disprove the simultaneous

$$\rho(b_X), \quad \|A_X - b_X\|_2 / \|b_X\|_2 \leq X^{o(1)} / J_X$$

requirements. A sharp lower bound on the best achievable error would be a publishable negative result.

7.2 Priority 2: determinant mass and equal-determinant coherence

Deliverable 2A: native-key census. Implement the TPC-84 streaming/external-sort schema with exact determinant and provenance keys. Report $M_X, D_X, E_{\text{trip}}, C_{\text{eq}}$, dominant-fiber mass, pairing-class totals, and the unpaired remainder under one normalization. This computation chooses the next theorem; it does not prove asymptotics.

Deliverable 2B: one growing lower bound. Target either:

$$M_X \geq X^{-o(1)} N_{0,X},$$

a positive same-scale lower bound for one literal pairing class with controlled remainder, or a rigorously quantified fixed-loss profile. If a fixed-power mass loss is proved instead, abandon the natural corner and recalculate the full TPC-88 ledger.

7.3 Priority 3: primitive affine arithmetic

TPC-90 shows that the next useful theorem is not another local root count. Expand one literal low-frequency or determinant block into the squarefree-supported Liouville correlation and seek a bound that is uniform in the growing slopes, intercepts, origins, masks, and prefixes. The preferred progression is:

- (i) one literal residual block with a fixed-power or arbitrary-log saving;
- (ii) a controlled family of square-divisor moduli with exact complementary return;
- (iii) the full multiplier-weighted low window or TPC-18 generic block; and
- (iv) complete outer reassembly.

At each stage, averaged shifts, frozen coefficients, and density-one ranges retain those qualifiers.

7.4 Priority 4: full-block reassembly, exact cover first

Export the TPC-16/TPC-18 hard packet to the TPC-91 incidence format. The first calculation is exact feasibility, preferably with rational or interval-certified arithmetic. There are three publishable outcomes:

- (a) an allowed exact cover with acceptable coefficient cost;
- (b) a dual residual or Farkas certificate that kills the proposed zero-residual block family; or
- (c) an explicit residual plus a new physical theorem controlling it.

Outcome (a) closes reassembly with zero residual. Outcome (b) does not kill the original TPC-18 residual route. Outcome (c) closes reassembly only at the rate actually proved. Primitive determinant dispersion remains a separate subsequent theorem.

7.5 Final merge audit

No branch is declared complete in isolation. Before any endpoint claim, rerun the following audit:

- (1) literal physical coefficients and one prescribed h_0 ;
- (2) actual support and all native keys;
- (3) one global atomic normalization;
- (4) canonical, complete representation with no deleted sectors;
- (5) the recorded determinant compatibility $\lambda_D \leq 2\eta_Z$; and
- (6) the independent strict physical budget $\Lambda_{\text{phys}} < 1/400$.

Then verify the TPC-18 reassembly (exact cover or a controlled physical residual) and primitive determinant input separately. Any duplicated loss is removed before the comparison.

Proposition 7.1 (Roadmap outcome trichotomy). *Every prioritized branch has three honest outcomes:*

- (i) a literal positive estimate, which advances the dependency graph at its proved level;
- (ii) an exact negative certificate, which permanently removes the specified route or family; or
- (iii) a conditional interface or numerical diagnostic, which is retained without promotion.

None of these labels implies a prime-pair theorem unless the entire remaining conjunction is proved.

Proof. The positive and negative certificates are the go/stop tests in [section 6](#); every other outcome lacks at least one promotion item in [proposition 2.4](#). The final statement follows because the completion conjunction remains unproved. $\square$

8 Audited conclusion

Theorem 8.1 (Ten-paper audit conclusion). *TPC-83 through TPC-91 establish the following exact chain of reductions:*

- (i) natural determinant energy is equivalent to natural literal post-bin mass;
- (ii) any determinant-energy lower bound with fixed loss $\lambda_D < 1$ must use asymptotically positive equal-determinant off-diagonal coherence;
- (iii) the distinguished zero mode is not removed by centering, conditional expectation, or content inversion;
- (iv) its coherent Fourier route reduces to a mesoscopic literal low-frequency window satisfying a rate-dependent filter condition (with $R_X = X^{133/600+\varepsilon} < q_X/2$ for fixed $0 < \varepsilon < 107/240$ in the second-order zero-loss corner);
- (v) logarithmic-to-ordinary transfer is measured by an exact signed Cesàro defect;
- (vi) in the natural-mass corner the flatness target is exactly relative cancellation $X^{o(1)}/J_X$;
- (vii) a surrogate transfers that target in the uniform worst case only with $O(J_X^{-1})$ relative accuracy in order, corresponding to $X^{o(1)}/J_X$ in the physical convention;
- (viii) primitive affine local squarefree obstructions are completely separated from the remaining Liouville sign correlation; and
- (ix) the zero-residual strengthening of full-block reassembly is an exact allowed synthesis equation with sharp dual obstruction certificates, while the original TPC-18 hypothesis still permits a separately controlled physical residual.

These are L0 closures and L1 interfaces. No paper in the block proves a new L2 growing fixed- h_0 arithmetic estimate.

Proof. Items (i)–(ix) are respectively the audited main results of TPC-83–TPC-91 summarized and cited in [sections 3](#) and [4](#). Each source states its remaining growing arithmetic hypothesis explicitly. $\square$

Positive progress. The program now knows the correct post-bin mass variable, the necessary coherence mechanism, the exact zero-mode target, a shorter Fourier window, the exact Tauberian defect, the sharp relative cancellation and surrogate scales, the complete primitive affine local factor, and a falsifiable zero-residual cover equation with a separate physical-residual alternative. These are not merely editorial refinements: each closes a required interface, strictly reduces a literal target, or removes an invalid route by exact counterexample.

Unresolved arithmetic. No lower bound for the actual M_X or D_X , upper bound for Z_X , growing affine Möbius/Liouville cancellation theorem, literal same-frame surrogate, physical full-block reassembly, or TPC-18 primitive determinant-dispersion theorem has been proved. The completion conjunction in [definition 5.1](#) remains open.

Independent ledgers. The recorded determinant bounds must satisfy

$$\lambda_D \leq 2\eta_Z.$$

Separately, all other physical fixed-power losses must satisfy

$$\Lambda_{\text{phys}} < 1/400.$$

Equality or excess stops the latter ledger. The same gain or loss is never entered in both.

Independent TPC-18 gates. Full-block reassembly and primitive determinant dispersion are different hypotheses. An exact cover is a sufficient zero-residual reassembly certificate but does not estimate its columns; column estimates prove neither exact coverage nor a physical residual bound. A nonzero residual remains admissible only with a separate theorem for its physical image.

Literal invariants. Every continuation retains one fixed nonzero h_0 , actual support, native keys, masks, prefixes, short fibers, polarizations, exceptional sectors, and one global post-aggregation normalization. A coarse-key, averaged-shift, fixed-form, or almost-all result remains at that scope until a literal transfer is proved.

Claim boundary. This paper is an L0/L1 audit and roadmap. It does not specialize h_0 to 2, prove a fixed-shift Hardy–Littlewood asymptotic, breach the sieve parity barrier, give a prime-pair lower bound, prove the twin-prime conjecture, or make any Riemann-hypothesis claim.

Next decision point. Run priorities one and two first: a literal zero-mode certificate and a post-bin mass/coherence theorem. Use TPC-90 as the arithmetic engine only after a specific physical block has been exported, and use TPC-91 to certify the zero-residual route or kill the proposed zero-residual family, while retaining the controlled-residual alternative. Stop any assembled route at physical loss $1/400$, but retain every exact negative certificate as a publishable boundary result.

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